

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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6 September 2026

Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity stepping down a squared-drift objective removes most of the objective available to it at the instant each update acts, and keeps the system coherent, while

driving within-cluster coupling far below between-cluster coupling—away from the structure it is asked to learn, and further from it than from almost every relabelling of the nodes; two controls sharing its step size, the second of them also its cumulative kernel deformation, do neither, and the same kernels are unremarkable against partitions of the same nodes drawn blind to frequency. Averaged over the intervals between updates the objective returns to its frozen level, and a bounded search over learning rates and update cadences finds coherence and lost alignment without a sustained reduction. Descending that functional does not produce representational learning: for this pairing of objective and template the optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms—which a sequence of equilibrated phase distributions cannot do, since a coupling increase, a diffusion decrease and a matched common drive share every stationary density.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination

7 (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible
8 interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch,
9 2015). Whether that mechanism is sufficient to support the collective state
10 such accounts require remains an empirical question.

11 Such an account has three requirements, and they order what follows.
12 Coupling must be strong enough to sustain coherent collective activity; the
13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery

30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset exponent
34 measured from a finite ramp is a function of ramp speed across the sweep of
35 Section 4, so an exponent fitted to a recovery trace measures the ramp at least
36 as much as the mechanism—and that holds for any analysis reading a critical
37 exponent off a driven transition, not only for ours. The third is a negative
38 result about plasticity: a coupling update stepping down a squared-drift ob-
39 jective removes most of the objective available to it at the instant it acts, and
40 preserves coherence, while moving *away* from the environmental structure it
41 is expected to learn—below almost every node relabelling, below all but a
42 twentieth of the partitions drawn blind to that structure, and beyond what
43 either of two controls does: a random update of the same step size, and that
44 same gradient step delivered to relabelled nodes, which reproduces its cumu-
45 lative kernel deformation. The reduction is a property of that instant rather
46 than of the run: averaged over the intervals between updates the objective
47 sits at the frozen arm’s level, and a bounded grid of learning rates and update
48 cadences, with criteria fixed in advance, contains no case combining a sus-
49 tained reduction with the coherence and the loss of alignment. Descending a
50 squared-drift functional does not produce representational learning, and the
51 direction of that failure follows from a conservation law rather than from the
52 particular run. The fourth is a machine-checked composition (Section 5) in

53 which finite information capacity, Landauer’s bound, a predictive dissipation
54 bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point
55 theorem yield one conditional theorem, with the eight hypotheses that com-
56 position consumes isolated and named; one witness assembled from distinct
57 toy components satisfies all eight at once, so their conjunction is not empty.

58 A prose table of assumptions would list the same eight items, so the fourth
59 contribution needs a word about method. What the Lean development adds
60 is that it makes the list non-negotiable. The characteristic failure in an
61 argument of this shape is not a false step but a silent change of subject:
62 a theorem about a stationary density read as a claim about a trajectory,
63 an inequality about a register read as a claim about learning, agreement of
64 phases read as compatibility of representations. Type-checking forbids each
65 of those substitutions, and an assumption needed to make a step go through
66 appears as a hypothesis in the term rather than as a word in a sentence. The
67 last of them is not merely forbidden but refuted by a constructed object: the
68 development carries a cover at a single common phase whose local densities
69 disagree where its patches meet, satisfying every other requirement of the
70 gluing theorem and failing compatibility alone (Section 5.2). That object
71 is what a prose table of assumptions cannot carry. Formalization does not
72 make the assumptions true; it makes them impossible to lose.

73 The physical model and the formal composition belong in one article be-
74 cause they are two halves of one commitment. The coupling-gated prediction
75 is worth testing under this framework, rather than as one more synchroniza-

76 tion model, only because the coherent state it produces is claimed to do
77 further work: to support a globally consistent representation and a reflexive
78 fixed point that we identify with phenomenal unity and a minimal self.

79 The distinction that does that work is between phases agreeing and de-
80 scriptions fitting together, and it is concrete. Take two cortical regions whose
81 receptive territories overlap, each carrying a probability distribution over
82 what occupies the shared part. Their oscillators can be locked to the same
83 phase at every instant while one region's distribution on the overlap places an
84 object left of fixation and the other places it right. The phases agree; the de-
85 scriptions do not, and no single distribution over the shared territory restricts
86 to both. A theory that identifies unity with synchrony cannot mark that dif-
87 ference, and so cannot separate a brain whose regions oscillate together from
88 one whose regions are about the same world. The sheaf condition marks
89 exactly it: local states glue to one global section when their restrictions to
90 shared regions coincide, and otherwise not at all. Coherence enters as the
91 physical condition under which compatibility can be maintained, not as a
92 substitute for it, and the self-prediction map then asks the further ques-
93 tion of whether the glued state contains an accurate model of itself. That
94 identification is a commitment, not a theorem, and Section 6 states it, ar-
95 gues for it, and says what would count against it. Its separation from the
96 physical tests is what makes both assessable: the cortical coupling hypothe-
97 sis and the recovery prediction can fail while the mathematical composition
98 stands, and the composition can stand while the identification is wrong. The

99 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
 100 Banach, 1922) therefore appear here for their biological composition, not for
 101 any new fundamental physical law.

102 **2 A coupling-gated model of cortical coher-** 103 **ence**

104 **2.1 Dynamics and stationary solutions**

105 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

106 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 107 K_{ij} has inverse-time units, with angles treated as dimensionless. The uni-
 108 form mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and
 109 phase diffusion are separate parameters; a frequency-spread width cannot be
 110 substituted for D when applying the identical-frequency stationary theory.

111 For identical frequencies in their rotating frame, the assumed stationary
 112 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

113 The I_n are modified Bessel functions, and ψ is the collective phase. Lean
 114 verifies normalization, the density's circular order parameter, and the self-
 115 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 116 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 117 ists with zero; the positive branch is continuous at threshold and strictly
 118 increasing with K .

119 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

120 This is a result about the stationary solution branch. Of the coherent branch
 121 we prove existence, uniqueness, continuous emergence at threshold and strict
 122 growth in K , but not that it is dynamically selected. The dynamical mean-
 123 field limit—propagation of chaos—is the missing link between the finite sys-
 124 tem and the stationary description (Sznitman, 1991), and no theorem in this
 125 development closes it. Exchange symmetry alone does not justify factoriza-
 126 tion of the joint law, particularly for an ensemble with an unpinned collective
 127 phase.

128 A spatial version replaces the finite sum by an integral against a kernel
 129 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-
 130 field strength is a kernel average, not a quantity that grows with the arbitrary
 131 measure of the substrate. Reduction of a heterogeneous spatial kernel to a
 132 scalar K is an additional modelling step, and the threshold in Eq. (2) does

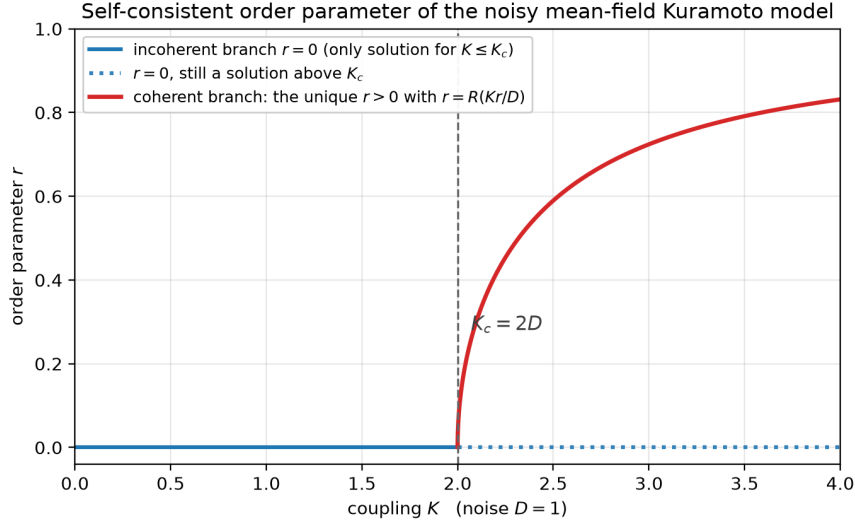


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

133 not automatically apply to every spatial or signed network.

134 2.2 Physical identification and calibration

135 Regular mesh refinement gives convergence of a discrete coupling-energy
136 quadrature to its continuum integral: scalar energy convergence, not conver-
137 gence of trajectories and not a construction of a physical kernel. The model
138 posits a continuum kernel and separately identifies its cortical realization
139 with endogenous electromagnetic coupling. The formal predicate requires
140 a jointly continuous kernel, a normalized substrate, a mean-field strength

141 equal to its spatial average, and no single site providing more than half of
 142 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
 143 model calibrated to cortex.

144 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
 145 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
 146 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 147 Fadden, 2020). These observations motivate an effective interaction without
 148 establishing its magnitude relative to phase diffusion, its spatial averaging
 149 rule, or its causal role in experience; changes in phase organization under
 150 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 151 coupling threshold.

152 The physical-unit conversion is the sharpest open limitation of the frame-
 153 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
 154 shift and oscillation frequency, is dimensionless. It cannot be compared di-
 155 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

156 requires a rate calibration γ , where s is the time shift per unit field and E
 157 the field amplitude. Supplemental Material sets out why γ is not recoverable
 158 from the other factors: it states how a per-cycle phase advance accumulates
 159 into a drift rate, and the choice that would let it cancel removes the very
 160 ratio the threshold test is about. The saved field-strength calculations are

conditional on this calibration and on the chosen spatial aggregation; they do not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase diffusion from the same physical system is necessary before the threshold becomes a quantitative biological claim; Section 3.2 makes that measurement the first item of the proposed experiment.

3 Recovery after awakening: a conditional prediction

Sleep-associated changes in extracellular space and clearance motivate a possible slow control of coupling (Xie et al., 2013). That geometry is measurable and its cortical values are known: the extracellular space occupies a volume fraction $\alpha \approx 0.2$ of tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al., 2013). It reaches Eq. (4) through the field amplitude E rather than through a further mechanism: below 10 kHz cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$, an order of magnitude set by the conductivity of saline diluted into that volume fraction (Barbour, 2017), so at fixed transmembrane current density a larger α yields a smaller extracellular field. We hypothesize that recovery after awakening includes a change in effective coupling $K(t)$, mediated by extracellular geometry, on a slower timescale than phase dynamics. The conductivity route fixes the sign that hypothesis predicts—contraction of the extracellular space on awakening raises K —so

182 the design of Section 3.2 can fail it in direction as well as in magnitude. It
 183 fixes nothing further: it is an effective-medium argument at fixed current
 184 density, and is silent on how a heterogeneous field aggregates into the scalar
 185 K of Eq. (2). Neither the monotonic relation between geometry and cou-
 186 pling nor its role in human sleep inertia follows from the cited physiology;
 187 the model requires both to be measured.

188 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 189 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

190 The ideal stationary description has a subthreshold interval followed by con-
 191 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 192 monotonic coupling alone does not determine a logistic or any unique sig-
 193 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 194 known. When coupling is inferred from the same coherence trace, this is not
 195 a parameter-free prediction of latency.

196 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 197 threshold, and a finite population carries nonzero fluctuating order below
 198 it. An observable onset therefore requires a predeclared threshold above the
 199 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 200 cusp. Section 4.1 quantifies the limitation.

201 **3.1 Distributional and temporal tests**

202 The stationary model makes a second prediction: phase concentration and
203 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
204 cability depends on the stationary-density assumption and the observation
205 model, and the relation is common to the whole von Mises family, so agree-
206 ment with it identifies neither ephaptic coupling nor this framework against
207 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
208 $r > 0$, so the limiting threshold ratio follows from the density relation instead
209 of testing it, and ratios near the finite-size floor are especially unstable.

210 The estimators must be independent, which is the constraint most easily
211 violated in practice. Estimating a by inverting the Bessel ratio from r would
212 enforce the relation being tested, and linear statistics on wrapped phases
213 estimate neither quantity.

214 **3.2 A protocol for the awakening experiment**

215 The requirements above define a design rather than a single measurement,
216 and the design is where the content of the hypothesis lives. An adequate test
217 needs all six of the following.

- 218 **1. Coupling and diffusion in the same units.** Estimate K and D
219 from the same cortical regime, in inverse-time units, separating phase
220 diffusion from quenched frequency spread and from observation noise.
221 Without this the threshold $K > 2D$ is not an evaluable statement

222 about cortex. A sequence of equilibrated phase distributions will not
223 supply it: a coupling increase, a diffusion decrease and a common drive
224 of matched strength give identical stationary densities window for win-
225 dow, so the estimate must reach the phase increments that carry the
226 rate, and telling an endogenous coupling from a common drive needs
227 a measured drive or a controlled perturbation on top of that (Supple-
228 mental Material).

229 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
230 an astrocytic signal is an indicator of the extracellular volume fraction
231 α , not a measurement of K . Its mapping to effective coupling must be
232 established separately, with its own uncertainty, and the conductivity
233 argument of Section 3 supplies only the expected sign of that mapping.

234 **3. Phase estimation at adequate spatial resolution.** At least 100
235 spatial samples per estimate. Estimate r by the circular resultant
236 length and a by an unweighted regression of log-density on the co-
237 sine of phase relative to the circular mean. With fewer sites, pooling
238 over time changes the estimand to a spatiotemporal distribution and
239 must be stated as such.

240 **4. An independent recovery endpoint.** Behavioral recovery measured
241 separately from the phase trace, so that latency is not defined by the
242 signal whose latency is at issue.

243 **5. A predeclared onset criterion.** A coherence threshold fixed above

244 the finite-size floor before unblinding, chosen in the knowledge that a
245 finite ramp smears the cusp.

246 **6. Model comparison under matched observation.** Coupling-gated
247 recovery compared against exponential, flexible sigmoidal and other
248 slow-recovery models under identical temporal filtering and observation
249 noise. Latency alone cannot distinguish the mechanism, because many
250 processes create delays; neither can a coherence trace alone, which the
251 mechanisms of item 1 share exactly.

252 The design fails informatively in three ways: failure of the stationary
253 phase-density relation over a sufficiently broad concentration range; absence
254 of the proposed coupling–geometry association; or a recovery trace better
255 explained by an alternative after finite-rate and observation effects are ac-
256 counted for. Two apparent failures are not failures. A linear apparent onset
257 is evidence against the adiabatic approximation only where that approxima-
258 tion is independently justified, and it does not falsify the field hypothesis
259 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
260 tion excludes stationary coexistence of two positive branches in this model,
261 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

262 4 Simulation results

263 The simulations characterize the finite-size, finite-rate regime that the sta-
264 tionary theory leaves open, and they bear on whether the prediction of Sec-

tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate methodological issues that also arise in analyses of driven transitions and dissipation-driven plasticity. Each is a finite run at its sampled parameters and not a general statement, and none discharges a hypothesis of the composition in Section 5. Each run uses explicit seeds and compact saved summaries, and the numerical values below are generated from those artifacts. The supplement gives parameters, estimators, failure criteria and additional panels. Identical-frequency runs use phase diffusion rather than a quenched frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

4.1 Ramp speed sets the apparent onset exponent

The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window. The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253, with a fitted delay exponent 0.443. Their apparent onset exponents run from 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the exact stationary branch sampled on the same coupling grids. At the fastest speed the ensemble mean never reaches the declared fit window, so the value the estimator returns there, 0.964, is a local log-slope of a delayed trajectory rather than an onset exponent.

The exponent recovered by fitting a driven transition is a joint function of the mechanism and the drive rate. An onset exponent read off a recovery

287 trace therefore says little about the underlying transition unless the ramp rate
 288 is independently known and the finite-size floor is declared, and that holds
 289 for any such analysis, not only for one conducted under this framework. The
 290 prediction of Section 3 is the first thing it constrains: the stationary square-
 291 root law cannot be imposed on an arbitrary recovery trace, and its absence
 292 is not by itself evidence against a coupling-gated mechanism. The specified
 293 deviation-based critical ramp speed is not bracketed by the sweep.

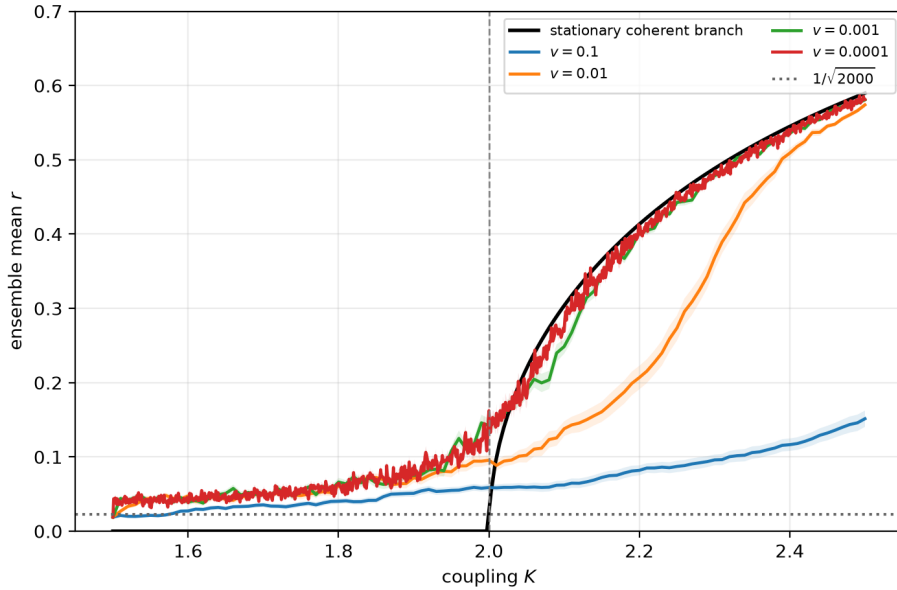


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

294 The dynamical-selection control shows escape toward a coherent branch

295 above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static
 296 value 0.68270. Refining the integration step by a factor of 4 at fixed dura-
 297 tion moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama
 298 bias; the last refinement still moves it by 0.00147, so the remaining -0.00102
 299 bounds the finite-size part rather than measuring it. The fitted growth-rate
 300 slope is 0.41505 against the theoretical $1/2$, over a window bounded below
 301 by the finite-size floor and above by the static branch; the same integrator
 302 bias enters that fit. The propagation-of-chaos diagnostics show decreasing
 303 subcritical dependence with increasing population size. In the supercritical
 304 unpinned ensemble, however, pair correlation remains substantial because
 305 realizations share a random collective orientation. This negative uncondi-
 306 tional result does not settle a conditional mean-field limit after removal of
 307 that orientation.

308 **4.2 Spatial coupling, frustration and calibration**

309 A periodic two-dimensional sheet with row-normalized exponential coupling
 310 remains coherent in the sampled empirical decay-length band under the de-
 311 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 312 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
 313 eter regime; fixing total coupling while changing range tests fragmentation,
 314 not whether cortex supplies that total coupling.

315 A row-balanced Dale-law network shows that row balance alone does not
 316 guarantee a floor-level incoherent baseline. In a separately specified weak-

synapse regime, a uniform modulatory term produces an operational coher-
 ence crossing inside the bracket $K_{\text{eff}}/D \in [1.958, 2.009]$, from a coupling grid
 refined to steps of 0.025 inside the one geometric step $[1.857, 2.110]$ that lo-
 cates it. The refined bracket still contains the model’s own mean-field thresh-
 old $K_c/D = 2$, so a tenfold finer grid does not separate the crossing from
 the threshold at this size. Signed coupling amplifies finite-size fluctuations
 in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which is the
 direction a resolved crossing would be displaced in; the refinement bounds
 that displacement by the width of the bracket rather than exhibiting it. The
 quantity K_{eff}/D is dimensionless; carrying it over to cortex needs the rate
 calibration γ of Section 2.2, which neither this sweep nor the spatial control
 supplies.

4.3 Squared-drift plasticity does not learn structure

The predictive dissipation bound of Section 5 bounds nonpredictive memory
 by dissipation, and it is tempting to read a system that lowers its dissipation
 as thereby coming to mirror the structure of its input. The bound licenses no
 such reading: a memory carrying no information at all satisfies it with room
 to spare, so lowering dissipation presses toward less representation rather
 than toward better-aligned representation. This control exhibits a system in
 which the pressure is not merely toward less but away.

337 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

338 It is a quadratic functional of an affine image of the kernel. The formalization
 339 constructs the bounded drift operator A and its adjoint on the relevant L^2
 340 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 341 decreases this objective at fixed phases. The objective is not generally a
 342 thermodynamic entropy-production rate, which requires probability-current
 343 information. Whether biological couplings follow this flow is an additional
 344 modelling question.

345 The joint control evolves noisy phases with clustered frequencies and
 346 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 347 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 348 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both
 349 endpoints contribute. Resource constraints and evolving phases remove the
 350 fixed-phase monotonicity guarantee. Each seed runs four arms on one phase-
 351 noise stream: the gradient step, a symmetric random step of the same Frobe-
 352 nius norm, that same gradient step under a random node relabelling, and a
 353 frozen kernel. The reported quantity is how much of the objective the update
 354 removes at the instant it acts, since diagnostics are sampled at the update
 355 cadence, immediately after each update: symmetric coupling cancels from
 356 the mean drift, which fixes a floor of 990 beneath the frozen arm's 1380.6063

357 to 1382.3448, and on that readout the gradient arm removes between 0.6773
 358 and 0.6914 of that headroom. The random arm of matched step size removes
 359 none of it, so the reduction belongs to the gradient direction. Matching the
 360 step size does not match the deformation it accumulates: successive isotropic
 361 steps cancel, leaving the random arm at a cumulative norm growth of 1.2879
 362 to 1.3056 against the gradient arm’s 12.4094 to 14.6958, so it barely enters
 363 the channel through which clip-and-rescale concentrates the fixed resource
 364 onto fewer edges. The permuted arm enters that channel. A relabelling is
 365 an isometry that leaves the multiset of step entries untouched, so the arm
 366 takes a step of the gradient’s size and of its shape on edges the gradient did
 367 not select: it reaches a norm growth of 11.4185 to 16.3296, bracketing the
 368 gradient arm’s, and a second-half order of 0.9184 to 0.9189, which lies inside
 369 the gradient arm’s range. Its descent fraction is -0.3319 to -0.1607 : at the
 370 gradient arm’s own deformation and coherence, a step that carries the gra-
 371 dient’s entries to the wrong edges raises the objective instead of lowering it.
 372 The learning rate 0.2 is the rate at which that readout falls furthest across a
 373 sweep from 0.001 to 0.4 evaluated on one of the three reported seeds, not a
 374 rate at which the update is known to converge.

375 Across three seeds, second-half order lies between 0.9167 and 0.9222: co-
 376 herence is maintained throughout. Alignment with the cluster template is
 377 read from the ratio of mean within-cluster to mean between-cluster coupling,
 378 which is unchanged by rescaling the kernel. That ratio starts between 0.9824
 379 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves

380 it at 0.9965 to 0.9997 and the equally deformed permuted arm at 0.7290 to
 381 1.2935, a range that contains the no-preference line and does not overlap the
 382 gradient arm's; against 2000 node relabellings the final kernel is further from
 383 the true template than from a fraction 0.9510 to 1.0000 of them (Fig. 3). The
 384 reference for that fraction is the frozen arm on the same seeds, which reaches
 385 0.6435 to 0.9110 without any update at all, and not a null median of one
 386 half; the increment the gradient step contributes is the difference between
 387 those ranges, which at three seeds is small enough that they nearly meet.
 388 The permuted arm reaches 0.2680 to 0.8410, so what carries the gradient
 389 arm above the frozen reference is the direction of the step and not the de-
 390 formation it accumulates. What the update moves against is that partition
 391 and not block structure at large. Read against 2000 partitions of the same
 392 group sizes drawn blind to frequency, at most a fraction 0.0495 of them give
 393 a lower within-over-between ratio than the frequency partition does on any
 394 seed, where the frozen arm's untouched kernel gives 0.0755 to 0.3480; and
 395 one such partition, interleaving the nodes so that each group draws equally
 396 from all three clusters, leaves the ratio at 0.8071 to 3.0223, inside that family
 397 and on both sides of the no-preference line. The update is opposed to the
 398 structure the environment carries rather than indifferent to structure. Two
 399 readouts do not carry the effect and we do not rest it on them. The initial-
 400 minus-final Frobenius distance to the template lies between -58.0962 and
 401 -48.1760 , so that distance grows; but the kernel norm grows by a factor of
 402 12.4094 to 14.6958 over the same interval, and a fixed total resource con-

403 strains the sum of the entries rather than the norm, so most of that growth
 404 is concentration of mass onto fewer edges rather than loss of alignment. The
 405 edge-level correlation with the template ends between -0.0362 and -0.0208 ,
 406 near its starting value: at this sparsity the edge-level variance swamps a con-
 407 trast that the block means resolve. The template is a covariance of shared
 408 latent cluster signals, not an input to the update or a measured covariance
 409 of the constant frequencies.

410 Most of the available objective is removed at the sampled instants, coher-
 411 ence is maintained, and within-cluster coupling falls to a fraction of between-
 412 cluster coupling. Coherence is therefore not a proxy for representational
 413 alignment in this regime, and a plateau in the objective establishes neither
 414 stationarity of the adaptive process nor a thermodynamic entropy-production
 415 rate.

416 What the update takes off the objective at the instant it acts, the interval
 417 that follows returns. Measured at every integration step and averaged over
 418 complete update intervals, a run at this rate and cadence sits at 0.998 and
 419 1.003 of its frozen arm across the two halves of its second half. A bounded
 420 study fixed in advance asks whether some other declared regime holds a re-
 421 duction while keeping coherence and losing alignment: 6 combinations of
 422 learning rate and update interval, each against a paired frozen arm, scored
 423 on complete intervals against criteria and a selection rule declared before
 424 the runs. None reaches the declared 5% sustained reduction, the best falling
 425 between 1.22% and 1.36% , while coherence and the loss of alignment ap-

426 pear across the grid. This regime therefore exhibits a coupling update that
 427 loses environmental structure while remaining coherent, and not one that
 428 sustains a lower objective; the second is what an argument from dissipa-
 429 tion minimization to representation would need, and a bounded grid that
 430 does not supply it does not show that nothing does. Nor is the direction
 431 of the failure accidental. Symmetric coupling conserves the mean drift, so
 432 minimizing $\sum_i v_i^2$ can only equalize drift across nodes, and equalizing drift
 433 rewards coupling between clusters of unlike frequency—the complement of
 434 a template that rewards coupling within clusters of like frequency. The ob-
 435 jective and the template are in this pairing opposed by construction, and
 436 that is the point: nothing about a system’s dissipation fixes which struc-
 437 ture its slow variables acquire. What generalizes beyond the pairing is that
 438 last sentence and not the sign of the effect, which is why the blind-partition
 439 reading matters—the update locates the environment’s own structure and
 440 moves against it, rather than degrading any structure a reader might score
 441 it against. What fails here is the objective rather than thermodynamically
 442 specified learning at large: an Ising model whose couplings are annealed de-
 443 grees of freedom, driven by external fields that carry the data, memorizes and
 444 generalizes under ordinary statistical-mechanical rules (Corberi et al., 2026).
 445 Those couplings equilibrate under a Hamiltonian containing the data; these
 446 descend a squared-drift functional whose stationary point the data never en-
 447 ters. Which functional the slow variables follow is what decides whether they
 448 acquire environmental structure.

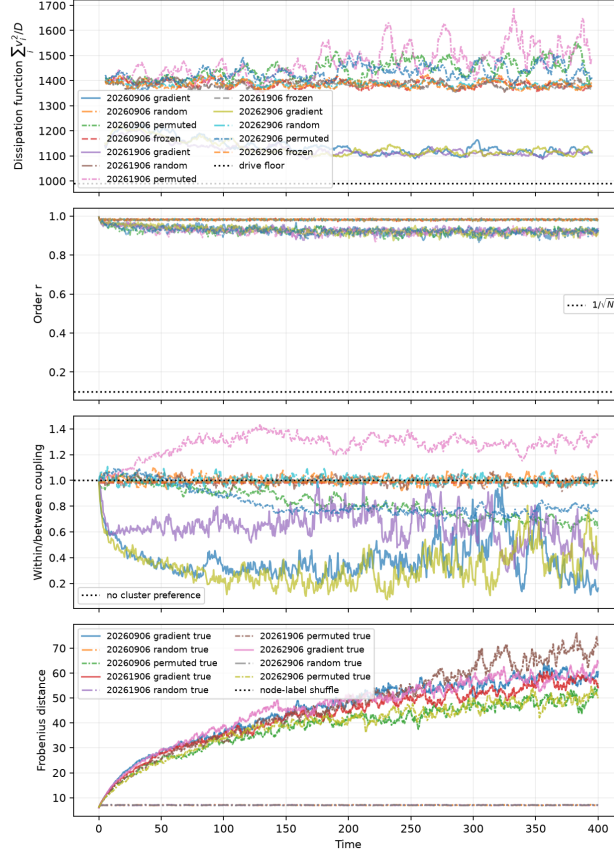


Figure 3: Joint fast/slow dynamics, four arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), the gradient step under a random node relabelling (dash-dot-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation, sampled just after each coupling update, against the drive floor the conserved mean drift imposes; only the gradient arm falls, and only on that readout. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line, the relabelled arm stays above the gradient arm at the same kernel deformation, and the remaining arms stay on the line. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

4.4 The estimation protocol demonstrated on public EEG

Item 3 of Section 3.2 specifies estimators whose behavior on real recordings is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa et al., 2025). The purpose is to show what the protocol produces and where it is fragile, not to test the awakening hypothesis, for which these data are the wrong design.

Two fragilities appear immediately. The implemented estimator pools 62 channels over a 100-ms window, so it estimates a spatiotemporal distribution rather than an instantaneous spatial one—a departure from the protocol forced by channel count, and one that any scalp implementation will face. Bipolar and common-reference montages give different residuals, making referencing a material analysis choice rather than a preprocessing detail.

The bipolar results are compatible with the Bessel relation over the observed concentration range. That range is too narrow to distinguish the curve from its linear approximation at the observed scatter, so the check is uninformative about the relation itself. An informative test needs a broader concentration range and explicit treatment of source mixing, finite samples and temporal pooling; intracranial or high-density recordings across a genuine state transition are the natural setting.

469 5 Thermodynamic constraints and the for- 470 mal composition

471 The synchronization model of Section 2 is one link of a longer conditional
472 construction, whose nine nodes are:

473 **n1** a finite phase space has bounded information capacity; **n2** a
474 broken symmetry forces a boundary, where the field leaves its vac-
475 uum manifold; **n3** the boundary erases, and erasure costs heat;
476 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
477 couplings coarse-grain to a continuum limit; **n6** a specified phys-
478 ical field realizes that limiting coupling; **n7** above threshold the
479 field carries a unique coherent order parameter; **n8** compatible
480 local states glue to a unique global section; **n9** that section is the
481 unique fixed point of a self-prediction map.

482 Node-level results are established or cited; what the composition consumes
483 are the eight implications between them. Figure 4 shows where they sit and
484 what kind of assumption each one is, and Table 1 states them, classified by
485 their role in the composition rather than by their plausibility. In particular,
486 finite capacity does not imply a symmetry-breaking boundary: **n1** and **n2** are
487 independent roots, even though the composition can accept an implication
488 between them as an argument.

489 5.1 Capacity, boundaries and predictive memory

490 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
491 equality only for the uniform distribution. A stream with more possible

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence drives a cover, supplied as a separate argument of the theorem, to an equilibrium reached by relaxation at coupling at least the named mean-field value. That its patches carry compatible local states is a field of the cover rather than a conjunct of this edge.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

492 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 493 a continuous field on a connected substrate that reaches distinct components
 494 of a disconnected vacuum manifold must leave that manifold somewhere,
 495 the simplest topological-defect construction (Kibble, 1976). Relating this
 496 locus to a particular finite information-processing register is the additional
 497 hypothesis E12 carries.

498 A non-surjective update on a finite register is non-injective. Under the
 499 specified system–bath model and thermal assumptions, erasure carries a Lan-
 500 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 501 update erases, and logical many-to-one processing must be distinguished from
 502 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 503 tion records the physical heat identity as an obligation of each model and
 504 supplies an explicit one-bit eraser satisfying it.

505 For a system state X_t driven by an environmental signal S_t , the predictive
 506 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

507 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 508 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 509 nonpredictive memory at a given dissipation budget. It does not specify
 510 a learning rule, maximize prediction, or require the internal representation
 511 to reproduce environmental geometry—the point the plasticity control of

512 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 513 ditionally requires nonpredictive information to be bounded by the entropy
 514 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 515 bound under that identification; it does not derive the identification.

516 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 517 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 518 tropy, not a general identification of relative entropy with physical entropy
 519 production. The chain uses a different technical sense of information at each
 520 step—a counting bound on a register, a thermodynamic heat cost, a predic-
 521 tive mutual information—and those senses are not interchangeable (Rosas
 522 et al., 2026). These distinctions prevent an informational inequality from
 523 being read as a derivation of environmental learning, Bayesian inference, or
 524 semantic content.

525 **5.2 Compatible states and reflexive fixed points**

526 A sheaf formalizes local descriptions and their restriction to smaller regions.
 527 If local sections agree on every overlap, the sheaf condition gives a unique
 528 global section. In the probability-sheaf construction, this is consistency of
 529 local density information, and it is not implied by agreement of oscillator
 530 phases: the development includes a counterexample with equal phases but
 531 incompatible densities on an overlap. Section 6 rests on this distinction,
 532 and the model excludes such configurations only through its compatibility
 533 hypothesis.

534 A thermodynamic cover couples this gluing construction to a phase con-
 535 figuration at a potential minimum. For the deterministic positive-coupling
 536 model, convergence to that minimum is proved under an explicit initial basin
 537 condition; other equilibria prevent an unconditional conclusion. The station-
 538 ary noisy order parameter, the deterministic relaxation, and the cover must
 539 therefore be connected by E78 rather than identified without a hypothesis.
 540 Neither field continuity nor synchronization alone guarantees a globally com-
 541 patible representation.

542 A reflexive extension equips global sections with a complete metric and a
 543 predictive map formed by restricting a state to a local encoding region and
 544 reading that encoding out. If the map contracts and fixes the section supplied
 545 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
 546 The contraction rate is assumed, and the equality between the glued section
 547 and the fixed point is part of E89. A coherent order parameter supplies the
 548 supercritical inequality required by the stated contraction bound; it does not
 549 construct the map or prove that cortical dynamics implements it.

550 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
 551 from two structures it also takes as arguments, a thermodynamic cover and
 552 a reflexive boundary. The eight edges are not the whole of what is assumed:
 553 the cover’s overlap-compatibility field—the load-bearing assumption of the
 554 unity-is-not-synchrony argument—and the boundary’s resonance condition
 555 are class fields, so they are consumed by the theorem without appearing
 556 among its edges. A single witness satisfies all eight at once, so their conjunc-

tion is not empty; it is assembled from a double well, a register, a uniform-grid mesh and a three-site cortex, which remain distinct toy systems rather than one mechanism. The project declares no additional Lean axioms beyond the foundational axioms used by Lean’s libraries; physical content resides in hypotheses and model fields. This checks the conditional mathematical composition, not the empirical truth of its premises. Detailed constructions, scope statements and counterexamples are retained in the supplement and source files.

6 What the construction commits to about consciousness

The composition terminates in a mathematical object, and we take a position on what that object is. The commitment has two clauses. *Unity*: the unique global section glued from compatible local states is the formal correlate of the unity of a conscious episode—the fact that a subject undergoes one experience rather than a collection of co-occurring local ones. *Selfhood*: the fixed point of the self-prediction map on that same section is the formal correlate of a minimal self, a state that is what it takes itself to be. Neither clause is a theorem, and no simulation here supports either. We argue for them, then say what would count against them.

Three features of the construction motivate the identification, and each distinguishes it from a bare synchrony measure.

578 First, compatibility rather than agreement. Oscillator phases can agree
579 everywhere while the local probabilistic descriptions they carry remain in-
580 compatible on their overlaps, and Section 5.2 exhibits that configuration
581 inside the formalization, not only as an illustration. If the unity of experi-
582 ence is a matter of local contents being about one world rather than merely
583 co-timed, then compatibility, not coherence, is the property to look for—and
584 coherence enters as the physical condition under which compatibility can be
585 maintained, which is what E78 asserts.

586 Second, uniqueness follows from the construction rather than from a sep-
587 arate selection principle. Accounts of conscious unity need something that
588 picks one experience from among the candidates a system could support;
589 integrated information theory postulates exclusion for this purpose (Tononi
590 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
591 whose local sections agree, the sheaf condition gives exactly one global sec-
592 tion, and given a contracting self-prediction map, Banach’s theorem gives
593 exactly one fixed point. The scope of that claim needs stating exactly, be-
594 cause the stronger reading is available and is not ours. Uniqueness is relative
595 to a cover and a map, and choosing them—which regions, which overlaps,
596 which encoding region, and so where the subject ends—is not supplied by ei-
597 ther theorem. It is what E78 and E89 carry. What the construction removes
598 is the need for an additional principle to select among the experiences a fixed
599 cover and map would otherwise admit; it does not remove the need to say
600 which cover and map a given subject has. That is a structural argument,

601 not evidence.

602 Third, the reflexive clause makes the self a condition rather than an addi-
603 tion. The predictive map restricts a global state to a region encoding it and
604 reads that encoding back out, so a fixed point is a state whose self-model is
605 accurate about the state it models. That condition is available only because
606 the state modelled is a single glued object: a system whose local descrip-
607 tions do not cohere has no such state for a self-model to be about. Unity
608 and selfhood are therefore one construction read at two levels, which is why
609 `UnifiedSelf` requires the fixed point to be the glued section itself rather
610 than an unrelated section.

611 The costs belong with the argument. The commitment does not address
612 why any physical condition is accompanied by experience; it is a proposal
613 about which physical state is the conscious one, not about why states of that
614 kind feel like anything. It is therefore open to the objection that a system
615 could satisfy every condition—compatible local states, a supercritical field,
616 an accurate self-model—while nothing is experienced. We have no reply to
617 that objection and claim none. The identification is also multiply realizable
618 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
619 or to electromagnetic coupling, so the substrate claim is carried entirely by
620 E56 and E67 and is empirical. No conclusion about artificial systems follows
621 from the formal results.

622 What could count against the commitment is dissociation. It predicts that
623 the two clauses and the presence of a unified conscious episode go together,

624 so a state in which local cortical descriptions are demonstrably compatible
625 and the field supercritical, while the subject reports no unified experience,
626 would tell against it; so would unified reported experience where compati-
627 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
628 states are the places to look, because they separate coherence from report
629 in ways ordinary waking does not. Such tests need an operational measure
630 of overlap compatibility on neural data, which this article does not supply;
631 constructing one is the main obstacle to making the commitment empirically
632 live rather than merely stated. Supplemental Material specifies what such an
633 observable would have to satisfy—a restriction map, a statistic with a null
634 under decoding noise, validation against configurations whose compatibility
635 is built rather than inferred—and the three ways it would fail silently, one
636 of which is reading compatibility off synchrony and so assuming away the
637 distinction being tested.

638 The physical tests of Sections 3 and 4 are independent of all of this. They
639 concern coupling, coherence and recovery, and can fail while the identification
640 stands, or succeed while it is wrong.

641 **7 Discussion**

642 The framework connects thermodynamic information constraints, collective
643 phase organization and compatible local representations, conditionally and
644 with its assumptions named. Its biological content is a coupling-gated re-

645 covery hypothesis with defined observables, a stated protocol, and failure
646 conditions declared in advance. Its formal content is a composition whose
647 eight gaps are exhibited rather than argued past, and which can be inhabited
648 by a single model. Its interpretive content is the identification of Section 6,
649 offered as a commitment with reasons.

650 The simulations are useful because they restrict the hypothesis rather
651 than support it. Finite ramps smear the stationary onset to the point where
652 a fitted onset exponent chiefly reports the ramp; an unpinned collective
653 phase makes independence diagnostics ambiguous in exactly the supercritical
654 regime the model cares about; and squared-drift plasticity maintains coher-
655 ence while degrading structural alignment. The second and third of these
656 are results about method and about a common intuition, and they transfer
657 beyond this framework.

658 The proposal differs in emphasis from accounts based on integrated in-
659 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
660 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
661 candidate interaction medium and a mathematical compatibility condition,
662 and, relative to a given cover and self-prediction map, it derives the unique-
663 ness those accounts postulate. It does not derive the exclusion principle of
664 integrated information, identify every broadcast with a field transition, or
665 derive a generative model from a dissipation bound. Continuous fields can
666 carry incompatible local descriptions and discrete models can admit sheaf
667 constructions, so continuity alone does not settle the combination problem;

668 compatibility is what this account offers in its place.

669 Two formal results bear on the substrate question, and both admit a
670 stronger reading than they support. At matched coupling resource, a fixed
671 wiring support that excludes a better-correlated pair can be suboptimal—a
672 statement about constrained optimization, not about biology. On an atom-
673 less substrate, a kernel supported on finitely many points contributes zero to
674 the chosen continuum coupling integral. That is a property of the measure
675 and the observable, and it cuts against this framework’s own coarse-graining
676 step as much as against anything else. It does not imply that digital hard-
677 ware cannot approximate a field or support consciousness, and no biological-
678 versus-artificial conclusion follows.

679 The empirical priorities follow directly from Section 3.2: estimate effec-
680 tive coupling and diffusion in consistent units, test the proposed relation
681 between coupling and extracellular geometry, and measure recovery against
682 an independently defined behavioral endpoint under matched model compar-
683 ison. A favorable concentration–coherence relation would support a statis-
684 tical description, not identify its physical cause; an unfavorable result in a
685 regime satisfying the stated assumptions would constrain that description.
686 Successful synchronization simulations cannot substitute for measurements
687 distinguishing ephaptic interactions from other coupling mechanisms.

688 The article therefore asks to be judged in three separable parts: a pre-
689 diction that can be tested, a composition that can be checked, and an iden-
690 tification that can be argued about. Keeping them separate is what makes

691 the whole assessable.

692 **Data and code availability**

693 The Lean development, simulation scripts, tests and saved numerical sum-
694 maries are maintained in the manuscript source repository. Publication sim-
695 ulation values are regenerated from saved summaries without rerunning pro-
696 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
697 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
698 in Supplemental Material. The source repository contains the reproducible
699 analysis and does not redistribute the raw recordings.

700 **Funding and competing interests**

701 This research received no external funding. The author declares no compet-
702 ing interests.

703 **Use of AI tools**

704 AI assistance to this work was substantial and is not separable by task.
705 Several coding assistants and chat interfaces were used in an interleaved way
706 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
707 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
708 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude

Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them they contributed to the Lean development, the simulation and analysis code, and the drafting and revision of both documents. Because the tools were used in combination and in alternation, no section, proof, figure or result is attributable to any one of them, and none is claimed to be. The author is responsible for all content, including everything a tool produced. Machine checking establishes the stated Lean results under their hypotheses; it does not validate biological interpretations or replace scientific review of generated code and prose.

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